

UDON THANI, Thailand

WMO: 483540

Lat: 17.377N

Lon: 102.809E

Elev: 178

StdP: 99.20

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.4	15.1	6.6	6.2	20.9	8.8	7.2	20.7	5.4	24.3	5.0	24.9	0.6	90	0.309

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	10.8	38.4	24.5	37.1	24.4	35.9	24.5	27.2	32.4	27.0	32.1	26.7	31.5	2.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	22.1	28.8	25.9	21.7	28.7	25.6	21.3	28.5	87.8	32.8	86.3	32.3	85.1	32.0	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years						
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max				
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.4	4.0	10.4	40.3	2.1	1.3	8.9	41.2	7.7	41.9	6.5	42.6	5.0	43.5		
			DB													
			WB	8.8	28.4	1.8	0.6	7.5	28.8	6.4	29.1	5.4	29.5	4.0	29.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.4	23.5	25.7	28.4	30.3	29.7	29.3	28.7	28.3	28.2	27.6	25.8	23.3
	DBStd	3.04	2.56	2.95	2.79	2.32	1.83	1.38	1.41	1.22	1.14	1.49	2.15	2.71
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	4	2	1	0	0	0	0	0	0	0	0	0	2
	CDD10.0	6348	419	439	570	609	610	578	579	568	546	544	473	414
	CDD18.3	3311	162	206	312	359	352	328	321	309	296	286	223	157
	CDH23.3	36403	1423	2126	3744	4686	4229	3868	3601	3324	3110	2931	2044	1318
	CDH26.7	15976	544	1024	2015	2615	2026	1673	1407	1194	1105	1129	794	449
Wind	WSAvg	1.4	1.2	1.3	1.4	1.4	1.4	1.4	1.5	1.5	1.3	1.3	1.2	1.3
Precipitation	PrecAvg	1465	7	18	46	77	207	221	228	279	251	95	14	5
	PrecMax	2418	71	115	208	219	420	656	536	573	642	294	126	58
	PrecMin	889	0	0	0	8	64	0	74	64	76	0	0	0
	PrecStd	300	15	21	48	49	89	94	98	111	117	70	25	11
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.2	37.0	39.3	40.5	38.8	36.2	35.9	34.5	34.2	34.1	34.7	33.4
		MCWB	21.9	21.9	23.1	24.9	26.0	26.3	26.3	26.4	26.2	25.6	24.1	23.8
	2%	DB	32.6	35.5	37.9	39.0	37.0	34.9	34.1	33.4	33.1	33.1	33.2	32.0
		MCWB	21.9	22.1	23.3	24.8	25.9	26.4	26.1	26.2	25.8	25.1	24.0	22.8
	5%	DB	31.2	34.1	36.5	37.6	35.4	33.9	33.1	32.4	32.3	32.2	32.0	30.8
		MCWB	21.5	22.5	23.4	24.9	25.9	26.2	25.9	25.9	25.6	24.6	23.5	22.0
	10%	DB	29.9	32.6	35.0	36.0	34.0	32.8	32.0	31.5	31.5	31.6	30.9	29.3
		MCWB	21.0	22.1	23.5	24.9	25.8	26.0	25.8	25.8	25.5	24.3	22.8	21.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	24.5	25.5	26.5	27.1	27.7	27.7	27.3	27.4	27.4	27.0	26.2	25.7
		MCDB	31.1	32.6	34.5	35.1	34.2	32.8	32.5	31.7	31.2	31.6	31.4	30.2
	2%	WB	23.5	24.5	25.6	26.5	27.1	27.2	26.8	26.9	26.9	26.3	25.3	23.9
		MCDB	29.7	31.6	33.4	34.5	33.4	32.1	31.4	31.1	30.7	30.5	30.0	29.3
	5%	WB	22.6	23.8	25.0	26.1	26.7	26.8	26.5	26.6	26.5	25.9	24.7	23.0
		MCDB	28.7	30.7	32.8	34.0	32.6	31.5	30.9	30.5	30.1	29.9	29.1	29.0
	10%	WB	21.7	23.1	24.5	25.6	26.3	26.5	26.2	26.2	26.2	25.5	24.1	22.2
		MCDB	27.9	29.7	31.8	33.1	31.8	30.9	30.2	29.8	29.6	29.2	28.5	27.9
Mean Daily Temperature Range	5% DB	MDBR	12.7	12.9	12.1	10.8	8.6	7.3	6.7	6.5	6.8	8.1	10.2	11.7
		MCDBR	13.9	14.5	14.3	12.9	10.7	8.8	8.3	7.9	8.0	8.9	10.9	12.2
		MCWBR	4.9	4.3	3.9	3.2	2.9	2.7	2.6	2.6	2.6	2.5	3.3	4.2
	5% WB	MCDBR	12.2	12.3	12.0	11.1	9.1	7.8	7.3	7.1	7.4	7.9	9.5	11.3
MCWBR		4.8	4.2	3.8	3.3	3.1	2.8	2.6	2.7	2.7	2.7	3.4	4.3	
Clear-Sky Solar Irradiance	taub		0.474	0.582	0.677	0.636	0.509	0.460	0.447	0.463	0.508	0.544	0.466	0.467
	taud		2.084	1.818	1.645	1.778	2.152	2.314	2.347	2.287	2.145	2.003	2.198	2.122
	Ebn at Noon		806	735	682	715	802	834	846	840	803	758	809	798
	Edh at Noon		155	212	260	228	155	130	126	135	155	174	138	146
All-Sky Solar Radiation	RadAvg		4.77	5.14	5.17	5.62	5.38	4.99	4.71	4.61	4.65	4.93	4.91	4.62
	RadStd		0.31	0.27	0.27	0.25	0.34	0.34	0.29	0.30	0.28	0.32	0.17	0.15

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+2	N/A	N/A
Regional (15 neighbors)	+0.27	N/A	N/A	N/A	N/A	N/A	N/A	+1	N/A	N/A

Nomenclature: See separate page